*Journal of Quantitative Analysis in Sports*

Manuscript 1340

# Playing Success and Local Market Size in Spanish Football League: Can Small Cities Dream of Winning Teams?

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# Playing Success and Local Market Size in Spanish Football League: Can Small Cities Dream of Winning Teams?

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**Abstract**

This article investigates the influence of local market size on playing success in professional Spanish football. In order to examine this relationship, several regressions were run for linear, beta and fractional logit models, using various proxies of local market size as explanatory variables. The evidence emerging from our analysis suggests that the advantage of big-market teams results in substantial dominance by a very small group of clubs. Nevertheless, this advantage does not actually prevent teams that are located in areas with a small market size from having at least regular opportunities for success. Furthermore, the econometric modelling and the evidence obtained provide a rational basis for supporting the idea of promoting the creation of a cross-national league in Europe (the European Superleague).

**KEYWORDS:** beta models, fractional logit models, market size, sports economics, team performance

**Author Notes:** The authors are grateful to the journal editor and two anonymous referees for their helpful comments. All remaining errors are our own. The authors also acknowledge financial support from Xunta de Galicia (Project 10SEC100041PR).

## I. INTRODUCTION

This article focuses on the role of local market size as a factor that determines the possibilities of a team's sporting success in the Spanish soccer league. This study has two main aims: firstly, to ascertain to what extent local market size is a relevant exogenous factor that determines sporting performance, and secondly, to test whether big-market soccer teams in Spain enjoy an insuperable advantage over clubs in small markets, preventing them from having regular opportunities for success.

It is generally accepted that location plays a relevant role in modern soccer, and that the geographic dependence of this sport is clearly visible at global, national and local level. The local geographical market size can be considered as an exogenous variable, as the team's location dictates the proximate population and the size of the market from which it draws its home attendance, its media audience and, as a result, its private revenue. Moreover, one of the main peculiarities of professional team sports is the expressive allegiance of fans to the teams. The degree of team loyalty in professional soccer implies that the market for team fans is highly segmented and has little mobility. Typically, fans feel attracted to teams based on their location, family affinity and/or sporting success, and teams are perceived as social assets (Gerrard, 2006).

The idea that team performance is affected by market size is nothing new. Previous research in the field of sports economics (see, e.g., Rottemberg, 1956; El Hodiri and Quirk, 1971; Walker, 1986) has shown that one consequence of the disparity in drawing potential among league teams is that there will be a tendency for teams located in strong-drawing areas to be, on average, stronger than those located in weak-drawing areas. Empirical evidence also suggests that market size is a determining factor in a club's revenue and its sporting performance (Noll, 1974; Quirk and Fort, 1992, Szymanski, 2003). Obviously, market size is not the only variable that influences sporting success, but it is reasonable to argue that this is a key factor. In particular, Zimbalist (1992) concludes that out of all of the explanatory variables in total team revenue equations, population is consistently the most significant.

This article is related to the above-mentioned literature in many aspects, but in particular focuses on the situation in Spain. The Spanish professional soccer league is one of the top five European soccer leagues, organized into several levels through a promotion and relegation system. All of the major professional Spanish soccer teams represent geographical areas, and they play as representatives of cities or regions. Recent professional soccer history in Spain seems to confirm that the town or city where the teams are based influences how the game is played on the field. Clubs are not free to choose their locations, and this constraint has serious consequences, as it gives some teams an inherent advantage due to the markets they serve. Professional teams competing in the first and second divisions are located in areas with very different drawing potential (a wide range of club revenues and city sizes) and, therefore, the critical question is to determine the scale of this big-city advantage. From this perspective, the analysis of the spatial distribution of successful sport teams within a geographical area is an interesting topic for study, especially in order to evaluate whether the long-term feasibility of the Spanish league is under serious threat as a result of its current dominance by a handful of teams.

In the light of these considerations, our aim is to obtain empirical evidence that confirms or rejects several widely-held beliefs regarding the influence of market size on playing success in professional Spanish soccer. In particular, we will attempt to answer the following, particularly relevant question: do big-city teams have some inherent advantage as a result of the markets they serve? How big is this advantage? Is this advantage the root of a possible serious competitive balance problem?

The rest of this article is structured as follows: in Section II, we review the theoretical models that predict a direct relationship between market size and sporting performance; in Sections III an IV we use an empirical approach to estimate the impact of market size on sporting success, and discuss the main results. Finally, we offer a summary of the main conclusions and policy implications derived from the previous analysis.

## II. Analytical Framework

The analytical models used in the literature on the economics of professional team sports shed light on the most critical factors in determining sporting success. In particular, depending on the assumptions about the main objective of teams' owners, two types of models can be identified. The first type are known as profit-maximizing models, and the second type include the models based on the utility maximization assumption.

The literature on sports economics has revealed different estimations depending on the hypothesis of profit-maximizing behavior by club owners. In North America, economists generally assume that clubs attempt to maximize their profits in the same way as any other enterprise. However, in Europe, greater emphasis has been placed on utility maximization, where the priority is winning matches. Nevertheless, although there are types of behavior that are consistent with utility maximization and inconsistent with profit maximization, from a theoretical standpoint, the existence of a direct relationship between team performance and market size is justified by both the profit and utility maximization models.

In a seminal paper, Scully (1974) develops a profit-maximizing model and notes that, "team revenues are directly related to the club's win percentage and to the size of the market from which it draws fans". When teams are located in cities with widely differing population levels, Scully's model implies cross-team variations in the demand for talent. It is profitable for teams in large cities to win more often than teams in small cities, because the marginal revenue generated by an additional win is higher in a larger city. Teams in large cities have a greater pool of fans to enjoy wins, and therefore wins should be more valuable for big-market teams than for small-market teams. In an open market for players, the best will gravitate towards the teams offering the highest salaries. Consequently, large cities will be able to pay higher salaries that teams in small markets cannot match.

In line with predictions of this kind, Burger and Walters (2003) present a general model of fan behavior and team performance, finding strong evidence that *ceteris paribus*, market size and team performance interactively affect marginal revenues, and therefore the willingness of teams to bid for playing talent. According to their estimates for Major League Baseball (MLB), teams in the largest markets obtain up to six times more marginal revenue from each additional victory than teams in the smallest markets. Indeed, the incentive to spend generously on acquiring players whose performance holds out a promise of more victories is great, but the ability and willingness of big-market teams to bid for skillful players will be far greater than that of their small-market rivals. The problem arises if these differences cannot be balanced, as teams are not free to move to areas where the marginal revenue per win is higher than in their initial location.

Furthermore, the aforementioned analysis also concludes that market size affects the marginal value of a win through the intensity and bandwagon channels. As a result, the revenue function for a team in a larger market will most likely have a greater slope than for a team in a smaller market, both below and above the win threshold needed to achieve contender status. The marginal value of a win will be greater for a contender than for a non-contender. Crossing the threshold from non-contender to contender status can raise the marginal value of extra wins. In other words, each team's revenue function will be kinked: its slope will increase significantly at the win threshold needed to achieve contender status. From this perspective, strict adherence to the dicta of profit maximization may result in some small-city teams having no chance of success. In a related study, Krautmann (2009) also finds strong evidence supporting the conclusion that it is really the higher marginal revenue of big-city teams that is the source of the dominance of large-market teams in baseball.

Sloane (1971) provides a basic analytical framework for models of utility maximization. If the owners put special emphasis on playing success, then they will be prepared to trade off some profit in order to secure additional playing success. If a club is a win maximizer, the only way the managers can maximize the winning percentage of their team is by hiring as many talents as can be afforded within the limits of the budget. Once the owner's minimum profits constraint has been met, most of the additional income will be spent on new players in order to improve playing success.

One of the main implications derived from a Walras equilibrium model is that the distribution of talent among clubs is more unequal if the aim of the club is to win rather than to make profits (Késenne, 2007). A big-market club in a win-maximizing league has more talents than the same club in a profit-maximizing league; inversely, a small-market club in a win-maximizing league has fewer talents than the same club in a profit-maximizing league. In other words, different club objectives may be expected to have a different impact on the number of talents hired by a club, and in line with the previous theoretical argument, big-market teams are more dominant in win-maximizing leagues than in profit-maximizing leagues, and the competition will be more unbalanced in the former.

Taking these considerations into account, the following causal sequence can be stated: the revenue earned by any team depends on the underlying drawing potential of its local market size. Therefore, given that teams are located in areas with different drawing potential, profit and utility incentives operating in a free competitive labor market will lead to a situation in which, on average, strong-drawing areas have strong teams, and weak-drawing areas have weak teams.

In order to address this question and to understand why it is especially relevant in a professional European league, in this case the Spanish soccer league (SSL), a few considerations should be made on two key differences between the professional leagues in North America and in Europe regarding club objectives and the organization of professional leagues.

First, Sloane (1971) argued that European club owners could be viewed as utility maximizers, with playing success as a major determining factor of utility, although profit is important for the club's financial stability in the long term. More recently, Primault and Rouger (1999) concluded that the economic objectives identified for teams differ between the US and Europe, due to the relegation and promotion system in Europe. In the particular case of the SSL, Garcia del Barrio and Szymanski (2009) provide some empirical support for the win maximization hypothesis. These authors conclude that player expenditure choices made by Spanish soccer clubs seem to closely approximate win maximization subject to a zero profit budget constraint; therefore, the SSL may be considered as a win-(utility) maximizing league.

Secondly, in North America, leagues are closed (i.e. they have a fixed number of members) and each league is essentially a monopolist franchise that serves to maximize its members' monopolistic profits. One of the direct consequences of this is that entry is rare and only granted by permission of the incumbents (the closed system); therefore, franchises can and do migrate. In contrast, the defining characteristic of the 'European' structure of professional soccer leagues is a system of promotion and relegation, effectively permitting entry on merit to all-comers (the open system). According to this system of relegation/promotion, a pre-determined number of teams at the bottom of a league or division are automatically relegated down to a lower level, and are replaced by the same number of teams who won promotion from the lower tier. Promotion and relegation have the effect of regularly rearranging the leagues according to the playing strengths of the different teams. In fact, there is promotion and relegation at every level of Spanish soccer, from the first division right down to the lowest level of amateur competition, which means that in theory, any Spanish soccer team is capable of reaching the uppermost level in Spanish soccer. Therefore, any individual or city that wants a top-league team can invest in the lower-level team that probably already exists there, and hope the team can advance to the top division.

Clubs may be sold privately to new owners at any time, but this does not happen often where clubs are based on community membership. Indeed, every sizable city or town at least has a semi-pro team (or teams) that will have probably secured the loyalty of the town's fan base, thus making the town unattractive to anyone interested in moving a team there, even if it plays in a higher division. Buying an existing top-flight club and moving it to the city is problematic, as the supporters of the town's original club are unlikely to change their allegiance to an interloper. As a result, clubs rarely relocate to other cities, and even private enterprises are reluctant to move.[1] Within this framework, anyone seeking to own a highly ranked club in their native city has to buy the local club as it stands, and then bring it up through the divisions, usually by hiring better talent.

In summary, as Szymanski (2003) argues, some of the main economic consequences of the promotion and relegation structure of soccer leagues are similar to the effects of open entry in any market. In particular, there is no credible threat of franchise relocation, since every city has at least one team with the potential to enter the major league (as long as it is prepared to invest in player talent).

## III. Empirical Analysis

In order to test the relationship between local market size and team performance in professional Spanish soccer, we have adapted Bradbury's empirical model

[1] The only case of team relocation in Spain took place at the end of the 2006-2007 season. Second division team *Ciudad de Murcia* was acquired by an investor from Granada, transferring it to the city and renaming it *Granada 74 CF*.

(Bradbury, 2007). This author studied the case of Major League Baseball (MLB), using a linear model, where the dependent variable was the number of wins, and the explanatory variable was the population of the metropolitan area, which worked as a proxy of the local market size. According to his results, obtained by OLS, the advantage of big-market teams over small-market teams appeared to be slight and virtually meaningless.

With the aim of comparing our results with the original Bradbury's specification, we estimate a model in which playing success is the dependent variable and the metropolitan area population is the measure of market size.

Our framework differs from those mentioned above in that firstly, we extend the analysis to another case study (a different sport and country); secondly, we use additional variables to measure the market size; and thirdly, we consider the case of a nonlinear relationship between market size and sporting performance rather than a linear relationship, as there are sound theoretical grounds to suppose that the linear functional form is not appropriate in this case. Finally, in order to isolate the individual effect of local market size on the performance of teams, in our empirical work the playing success index is regressed on alternative measures of market size including other control variables.

### *Data and Variable Definitions*

The dependent variable for this study is *playing success*. A long-term sports performance index was used as a proxy for this variable, calculated using data from a series of teams playing in the Spanish soccer league both in the first and second division, from the 1995-1996 season to the 2006-2007 season. As our interest lies in the structural influences on the performance of soccer teams rather than short-term influences, we have used a ten-year reference period in order to obtain an index of team strength. Another reason for choosing this period was that during these seasons the same number of teams played in the first and second division championship taken as a whole, while prior to 1995 the number of teams varied. This choice allowed us to construct our database, which consists of sixty-one teams.

The procedure used to obtain a *playing success index* for each team was as follows: firstly, for each season, we ordered the final league classification tables from the last team in the second division to the first in the first division, and then we sequentially assigned a number from 0 to 41 to each team. This means that the worst team had 0 points, while the best had 41 points, i.e. depending on how well or badly a team performed, it was assigned a higher or lower number of points. Secondly, for each team we added the points assigned according to the previous step for each season from 1995 to 2007. Finally, the score obtained was divided by 492, which is the maximum number of points that a soccer club could obtain in these circumstances (41 points/season x 12 seasons between 1995 and 2007 = 492 points); in this way, we obtained a *playing success index*, which is a proportion observed in the interval from 0 (the team that was the worst in every season) to 1 (the team that was the best in every season).

The main reason for using this dependent variable instead of winning percentage is that according to Hall *et al.* (2002), the promotion and relegation structure of league soccer renders the interdivisional comparison of winning percentages meaningless. Therefore, some other measure of playing success needs to be adopted. Although final league ranking may be used directly as the dependent variable, this measure is not very useful for our purposes, as it shows an inverse relationship with most of the variables we use. This is the reason why, in line with most authors, we work with a modified variable that is a monotone decreasing function of ranking (Szymanski and Kuypers, 1999; Dobson and Goddard, 2001).

With regard to the independent variable of *local market size*, several proxies were used, expressed as logarithms[2]: *adjusted provincial population*, *adjusted town population*, *market share* and *potential market*.

For the population indicators, we used data taken from the last available Spanish Census (2001). The relevant population was determined by the inhabitants in the province[3] (provincial population) or the city (town population) hosting a team. For the town population variable, we used linked population data, defined as the group of people who can be included in the census (i.e. who are normally resident in Spain) who have some kind of habitual link with the municipality in question, either because they live, work or study there, or because, although it is not their usual place of residence, they spend certain periods of time there (summer holidays, long weekends, etc.).

In cases where there was more than one team in the same geographical local market, we came across the intrinsic difficulty of assigning the correct proportion of the market size to the various teams. In order to assign the value of the population variable (*adjusted provincial population* and *adjusted town population*) to each team, the following considerations were taken into account.

On the one hand, the population share was equal to the provincial population when there was only one club in that province. When a province had more than one team, the population was divided equally between these clubs in

[2] Most of the proxies of the market size are rather right skewed, so it is better to add these variables on the log scale. This offers two advantages: firstly, it allows us to avoid including quadratic terms that are difficult to interpret, and secondly, when we consider changes in market size, it seems to be more common to think in terms of percentage changes rather than differences, and percentage changes are additive changes on the log scale.

[3] Within this context, Spanish provinces are comparable to the Metropolitan Statistical Areas (MSAs) in other countries (e.g. the USA).

cases where each club was in the same division.[4] Where clubs in the same province were in different divisions, the population was divided between them on the basis of the proportions of total attendances accruing to each division nationwide during the period of reference.[5] On the other hand, in the case of the town population, the population was divided according to average attendance at the respective club stadiums in the last seven seasons.

With regard to the *market share* and the *potential market* variables, the data source is the Spanish savings bank La Caixa (2007), and where necessary, the sharing criteria mentioned above for the case of cities were applied. The *market share* is an index that captures the comparative consumer capacity of the municipalities, obtained using a model equivalent to an average of index numbers of the six following variables: population, number of telephone land lines, cars, lorries and vans, bank offices and retail commercial activities. These index numbers express the share (as a rate per 100 000) that corresponds to each town on a national basis of 100 000 units. The *potential market* is a variable that measures the volume of annual retail sales for a municipality in 2006.[6]

In order to prevent the models from having an omitted variable bias, we use a number of control variables. In particular, we tried to include some additional explanatory variables that other authors have used in previous sports literature[7] when analyzing determinants of team performance, namely team payroll, managerial change and the age of the team. Team payroll can be considered as a proxy of team quality, managerial change works as a proxy of management efficiency and finally the age of the team takes into account the fact that teams with longer histories may have made greater development efforts.

From a theoretical perspective, a strong relationship can be expected between how well a team performs on the field and the average salary that the team pays to its players. Unfortunately, due to a lack of data on the payrolls of Spanish soccer teams, we were forced to use their operating revenues as a proxy. In our case, the choice of these revenues as a proxy of the teams' payrolls would be justified because according to data for the 2006-07 season, the payroll accounts for an average of around 75% of the total expenditure. In addition, the ratio of operating expenses and operating income is 1.08.

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[4] Teams that played most of the seasons between 1995 and 2007 were considered as 'First Division teams' and otherwise as 'Second Division' teams.

[5] If a team had usually played in the second division during this period, then we assigned it one third of the total population of the province. The rationale for this assigning criterion is that on average, attendance to second division fixtures in Spain is one third of the attendance for first division games.

[6] The year-on-year variation of this type of data, especially referring to population, is not considerable. As a result, we have chosen to focus our analysis on data from a single year.

[7] See, for example, Audas et al. (2002) and Hall et al. (2002).

However, in the case of Spanish soccer, a correlation analysis between this variable (i.e. operating revenues) and the different measures of market size revealed that they were closely related. Therefore, if we incorporate this variable into the model, this would lead to a problem of multicollinearity (with the subsequent risk of misspecification of the regression). The same problem would arise if we include other variables such as income, stadium size, number of years in current division, etc. Nevertheless, this kind of evidence suggests that the inclusion of the team payroll and local market size as predictors of a team's performance can be, to some extent, redundant.

Regarding the relationship between managerial efficiency and team performance, theory yields ambiguous predictions for the effects of head coach change on team performance and empirical evidence is also mixed. A review of the data on managerial changes in Spanish professional soccer from 1995-96 to 2006-2007 shows that coach replacement is common both between and during seasons.

Taking these considerations into account and in order to identify the long-term effects of managerial changes on team performance, we include the average number of coaches per season that each team had over the entire sample period in our regressions as an additional explanatory variable.

Finally, on the suggestion of an anonymous referee, we control for the age of the team. Some researchers have examined the impact and influence of variables such as the age of the franchise on the support and attendance of fans (Siegfried and Eisenberg, 1980; Depken, 2001) and the value of the franchise (Miller, 2007). Most of these studies found a statistically significant impact of the franchise age on these variables. In our case, in order to control for the age of the team we test whether teams with longer histories have made greater market development efforts. With this aim in mind, we include the year when each club was founded as an explanatory variable.

*Model Specifications*

We tested several alternatives, considering the four proxies for local market size, including the variables *managerial change* and *foundation year* as team-specific regressors. Two different regressions were run for each case:

- Model 1: beta model estimated by Maximum Likelihood (ML).
- Model 2: fractional logit model estimated by Quasi-Maximum Likelihood (QML).

Model 1 assumes that the playing success follows a beta distribution:

$$f(y/\mu,\Phi) = \frac{1}{B(\mu,\Phi)} y^{\,\mu\cdot\Phi-1}(y-1)^{(1-\mu)\,\Phi-1} \tag{1}$$

where μ and Φ are, respectively, one location and one scale parameter ($\mu > 0$, $\Phi > 0$), $B(\cdot)$ is the beta function and $0 < y < 1$. This models heteroskedasticity in such a way that the variance is greatest when the average proportion is near 0.5, and models the mean allowing different teams to have different μ depending on the values of the explanatory variables, using the logistic transformation to ensure that μ remains between 0 and 1.

Model 2 is an alternative proposed by Papke and Wooldridge (1996) to overcome the fact that although the implied variance of the dependent variable in the beta modeling makes sense, it is still an assumption and some people consider it to be too restrictive. Fractional logit can handle proportions of exactly 0 or 1, unlike beta distribution.

The reason why we chose these models for our study responds to the considerations put forward by Kieschnick and McCullough (2003). These authors, after a detailed analysis of the pros and cons of different alternative regression models for proportions, recommend researchers to use either the beta regression model or the fractional logit model, which prove to be the best, and they also claim that their case studies suggest that the larger the sample, the closer the results of these models will be, although in the case of small samples, the beta regression is preferred.

It should be emphasized that Bradbury's original model is not suitable in our circumstances, i.e. having a proportion as a dependent variable. As Kieschnick and McCullough (2003) point out, although researchers most frequently estimate the parameters of a linear regression model for proportions – variables observed either on (0, 1) or [0, 1] – using OLS, this approach contravenes two conditions: the conditional expectation function must be nonlinear since it maps onto a bounded interval; and its variance must be heteroskedastic since the variance will approach zero as the mean approaches either boundary point. Due to these weaknesses, some problems arise: impossible predictions (out of the range 0-1), non-normal errors, heteroskedasticity and nonlinear effects (Buis, 2006).

*Estimation Results*

In Table 1, we report the results of fitting the various regression models described above to the data, also described above. Apart from the results, which include the coefficients estimates and the standard errors of these estimates (in brackets), we indicate the significance level of each estimate, as well as the value of Akaike's

Information Criterion (AIC), as it is the most widely used and accepted criterion of model selection.

Table 1 shows that different inferences regarding the statistical significance of the regressors depend on the choice of regression model. The results are not surprising and in line with theory, local market size appears to have a positive and very significant[8] effect on the sporting performance of Spanish soccer clubs, irrespective of the chosen variable (*adjusted provincial population*, *adjusted town population*, *market share* or *potential market*) and the econometric model (beta or fractional logit).

*Managerial change* shows a negative and statistically significant effect (level of 1-5%, depending on the model) on team performance. The data suggest that the length of coaching tenure is a significant variable affecting that performance. This is consistent with Grusky's hypothesis (Grusky, 1963) that the higher the turnover rate, the poorer the team performance. Koning (2003) finds that the performance of a team does not always improve when a coach is fired. In some cases, new coaches perform worse than their predecessors who were fired. The impact of managerial skill on the field is equally important. Kahn (1993) reaches the conclusion that baseball managers substantially contribute to the process of winning, that these managers improve with tenure on a given team, and that players appear to respond more to better managers.

*Foundation year* has a negative and very significant[7] impact on team performance. Consequently, there is an 'age effect'. This implies a disadvantage for the 'youngest' teams in comparison to those with a long life in sports competition. This result suggests that the longer a team has resided in its host city, the more likely it is to be stronger. This may reflect an accumulation of goodwill toward the team based on team history within the host city, and also an indication of the building of fan loyalty for the team.

According to AIC, Model 1 is the best for all the variables (*adjusted provincial population*, *adjusted town population*, *market share* and *potential market*). Furthermore, the *adjusted town population* is the variable which proves to have the best explanatory power. The worst AIC is for Model 2, using the variable *adjusted provincial population*.

Since Models 1 and 2 are nonlinear, we cannot simply take coefficient estimates across regressions in order to obtain a sense of the estimates of the effect of local market size on team performance. For this purpose, we have to use the so-called marginal effects ($\partial y/\partial x_i$), which help us understand the impact of each covariate. Table 2 reports the marginal effects of the beta regression models with respect to the different regressors, evaluated at their sample means. The interpretation of the marginal effects is straightforward. For instance, considering

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[8] A significance level of 1% in all the cases.

**Table 1: The influence of local market size on team's performance**

| **Adjusted provincial population** | | |
|---|---|---|
| | Model 1 | Model 2 |
| Constant | 23.7527** (9.9516) | 30.0066*** (9.6236) |
| Market Size | 0.7653*** (0.1854) | 0.9154*** (0.2159) |
| Managerial change | (-0.8453)** (0.3355) | (-1.2262)*** (0.2947) |
| Foundation year | (-0.0144)*** (0.0050) | (-0.0179)*** (0.0049) |
| AIC | (-0.6948) | 0.9237 |
| **Adjusted town population** | | |
| | Model 1 | Model 2 |
| Constant | 25.0343*** (8.7848) | 27.2514*** (9.5369) |
| Market Size | 0.9703*** (0.1798) | 1.0709*** (0.2365) |
| Managerial change | (-0.6740)** (0.3010) | (-1.0905)*** (0.3055) |
| Foundation year | (-0.0154)*** (0.0045) | (-0.0165)*** (0.0048) |
| AIC | (-0.9362) | 0.8844 |
| **Market share** | | |
| | Model 1 | Model 2 |
| Constant | 30.3625*** (8.8157) | 31.0147*** (9.9708) |
| Market Size | 0.8727*** (0.1669) | 1.0257*** (0.2148) |
| Managerial change | (-0.7794)** (0.3146) | (-1.1951)*** (0.2996) |
| Foundation year | (-0.0181)*** (0.0046) | (-0.0186)*** (0.0051) |
| AIC | (-0.8872) | 0.8879 |
| **Potential market** | | |
| | Model 1 | Model 2 |
| Constant | 30.3827*** (8.7717) | 28.5352*** (10.6354) |
| Market Size | 0.8526*** (0.1635) | 1.0485*** (0.1900) |
| Managerial change | (-0.7700)** (0.3126) | (-1.2141)*** (0.3238) |
| Foundation year | (-0.0184)*** (0.0045) | (-0.0177)*** (0.0055) |
| AIC | (-0.9082) | 0.8760 |

*Notes:*
** means statistic significance at the 5% level.
*** means statistic significance at the 1% level.

the case of the results for the *adjusted town population*, we could claim that, on average: (i) an increase of 1% in market size would raise teams' performance by around 0.21 %; (ii) each change of coach is associated with a performance loss of almost 0.15%; (iii) a team founded, for example, ten years earlier than another would have a performance advantage of 0.03 %.

Table 2 suggests that there is no significant difference between regression models in their estimates of the marginal effects of the various regressors when evaluated at the same means of the regressors. Despite these results, we would draw attention to the fact that the models could provide quite different estimates of the marginal effects of different regressors, as these regressors are evaluated at points other than their means. This fact should not be overlooked when evaluating the significance or implications of our regression results.

**Table 2: Local market size and teams' performance. Estimates of the marginal effects (M. E.) of regressors (beta model)**

| | |
|---|---|
| **Adjusted provincial population** | |
| M. E. of Market size | 0.1705*** |
| M. E. of Managerial change | (-0.1883)** |
| M. E. of Foundation year | (-0.0032)*** |
| **Adjusted town population** | |
| M. E. of Market size | 0.2113*** |
| M. E. of Managerial change | (-0.1468)** |
| M. E. of Foundation year | (-0.0034)*** |
| **Market share** | |
| M. E. of Market size | 0.1903*** |
| M. E. of Managerial change | (-0.1699)** |
| M. E. of Foundation year | (-0.0040)*** |
| **Potential market** | |
| M. E. of Market size | 0.1849*** |
| M. E. of Managerial change | (-0.1670)** |
| M. E. of Foundation year | (-0.0040)*** |

*Notes:*
** means statistic significance at the 5% level.
*** means statistic significance at the 1% level.

## IV. DISCUSSION OF RESULTS

Taking into account the results of the previous regressions, starting from the specification of Model 1 for the variable *adjusted town population*, we found it reasonable and interesting to carry out an analysis that was similar to some extent to Bradbury's exercise for the Spanish league. In doing so, in order to isolate the effect of the market size indicator, we suppose that in this model: (i) the variable *managerial change* is constant and equal to 1 for all the teams (i.e., each team had just one manager during each season) and (ii) the variable *foundation year* is constant and equal to 1930 (sample average). As a result, the regression procedure generates two basic indicators: *Predicted performance* and *performance above/below the predicted level*.

*Predicted performance* is the sporting performance that teams should achieve, solely based on their local market size. This was obtained from the beta model (1) for adjusted town population, making the assumptions explained in the former paragraph for the variables *managerial change* and *foundation year*.

The *performance above/below the predicted level* is the difference between actual observed sporting performance and *predicted performance*:

$$PABP = OP - PP \tag{2}$$

where *PABP* is the *performance above/below the predicted level*, *OP* is the observed performance and *PP* is the predicted performance.

Table 3 lists Spanish soccer teams ranked according to *performance above/below predicted* (last column).

The data on *observed performance* reveal the dominance of big-market teams, that is, teams located in a large city combined with a large fan base (i.e. Barcelona, Real Madrid, Valencia). Within the domestic championship, there is a considerable gap between a handful of top teams and the rest of the league.[9]

Using the value of a team *predicted performance* (determined by its local market size) as a proxy for quantifying fans' success expectations, a new way to measure team performance in relative terms can be defined: *observed performance above predicted performance*. The comparison between *predicted performance* and *observed performance* allows us to conclude that there are several teams with similar predicted performance yet highly differing observed performance.[10] Figure 1 shows the scatterplot of the *predicted performance* (PP) *vs. performance above/below the predicted level* (PABP). Each of the points of the graph represents a team, and its distance from the horizontal line at zero indicates how far this team is (above or below) from the level of playing success predicted according to its local market size.

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[9] This result is consistent with the hypothesis that both the deregulation of the player transfer market (following the Bosman ruling) and the deregulation of the market for sports rights have contributed to the dominance of successful clubs located in large metropolitan areas.

[10] For an in-depth analysis of relative efficiency of soccer clubs in the Spanish Soccer League, see Pestana et al. (2009). These authors propose a framework for the comparative evaluation of Spanish soccer club performance using a random frontier model.

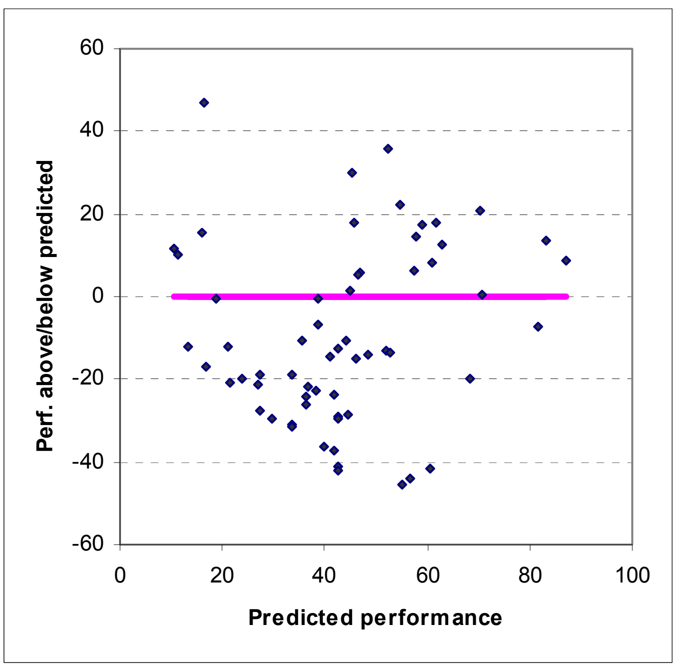
**Table 3: Predicted performance and performance above (+) or below (-) predicted of the Spanish football teams (Model 1, *adjust. town population*).**

| Team | Observed performance | Predicted performance | Perf. above/ below predicted |
|---|---|---|---|
| Villarreal | 63,01 | 16,24 | 46,77 |
| Deportivo A Coruña | 88,21 | 52,27 | 35,94 |
| Real Sociedad | 75,41 | 45,35 | 30,06 |
| Celta Vigo | 77,03 | 54,78 | 22,26 |
| Valencia | 91,46 | 70,44 | 21,02 |
| Rácing Santander | 63,62 | 45,56 | 18,06 |
| Athletic Bilbao | 79,27 | 61,57 | 17,70 |
| Real Betis | 76,42 | 58,85 | 17,57 |
| Numancia | 31,30 | 16,05 | 15,25 |
| Mallorca | 72,15 | 57,68 | 14,48 |
| FC Barcelona | 96,54 | 83,04 | 13,51 |
| Español | 75,20 | 62,72 | 12,48 |
| Extremadura | 22,15 | 10,68 | 11,48 |
| Eibar | 21,75 | 11,49 | 10,26 |
| Real Madrid | 95,93 | 87,24 | 8,70 |
| Sevilla | 69,51 | 61,05 | 8,47 |
| Valladolid | 63,62 | 57,45 | 6,17 |
| Deportivo Alavés | 52,64 | 46,77 | 5,87 |
| Osasuna | 51,83 | 46,59 | 5,24 |
| Tenerife | 46,54 | 45,04 | 1,51 |
| Zaragoza | 71,34 | 70,72 | 0,62 |
| Mérida | 18,50 | 18,76 | -0,26 |
| Albacete | 38,21 | 38,66 | -0,45 |
| Recreativo Huelva | 32,11 | 38,80 | -6,69 |
| Atlético Madrid | 74,59 | 81,83 | -7,24 |
| Salamanca | 33,74 | 44,26 | -10,52 |
| Compostela | 24,80 | 35,36 | -10,56 |
| Écija | 1,42 | 13,31 | -11,89 |
| Polideportivo Ejido | 8,94 | 21,27 | -12,33 |
| Las Palmas | 29,88 | 42,65 | -12,78 |
| Sporting Gijón | 39,02 | 52,07 | -13,04 |
| Rayo Vallecano | 39,23 | 52,84 | -13,61 |
| Oviedo | 34,55 | 48,53 | -13,98 |
| Getafe | 26,63 | 41,06 | -14,44 |
| Levante | 30,89 | 46,04 | -15,14 |
| Vecindario | 0,00 | 16,91 | -16,91 |
| Toledo | 8,54 | 27,29 | -18,75 |
| Lleida | 14,84 | 33,73 | -18,89 |
| Málaga | 48,78 | 68,38 | -19,60 |
| Lorca Deportiva | 3,66 | 23,71 | -20,05 |
| Ponferradina | 0,41 | 21,44 | -21,03 |

## Table 3 (Continued)

| Team | Observed performance | Predicted performance | Perf. above/ below predicted |
|---|---|---|---|
| Rácing Ferrol | 5,89 | 27,00 | -21,10 |
| Badajoz | 15,04 | 36,59 | -21,55 |
| Cádiz | 15,45 | 38,17 | -22,73 |
| Xerez Deportivo | 18,29 | 41,97 | -23,68 |
| Nástic Tarragona | 11,99 | 36,18 | -24,19 |
| Logroñés | 10,16 | 36,24 | -26,07 |
| Pontevedra | 0,00 | 27,43 | -27,43 |
| Elche | 15,65 | 44,42 | -28,77 |
| Leganés | 13,62 | 42,69 | -29,07 |
| Almería | 13,01 | 42,62 | -29,61 |
| Algeciras | 0,00 | 29,66 | -29,66 |
| Jaén | 2,85 | 33,63 | -30,79 |
| Ourense | 2,24 | 33,65 | -31,41 |
| Castellón | 3,66 | 40,03 | -36,37 |
| Terrassa | 4,47 | 41,90 | -37,43 |
| Burgos | 1,22 | 42,50 | -41,28 |
| Murcia | 19,11 | 60,55 | -41,44 |
| Univ. Las Palmas | 0,41 | 42,65 | -42,25 |
| Hércules | 12,80 | 56,77 | -43,97 |
| Córdoba | 9,35 | 54,96 | -45,61 |

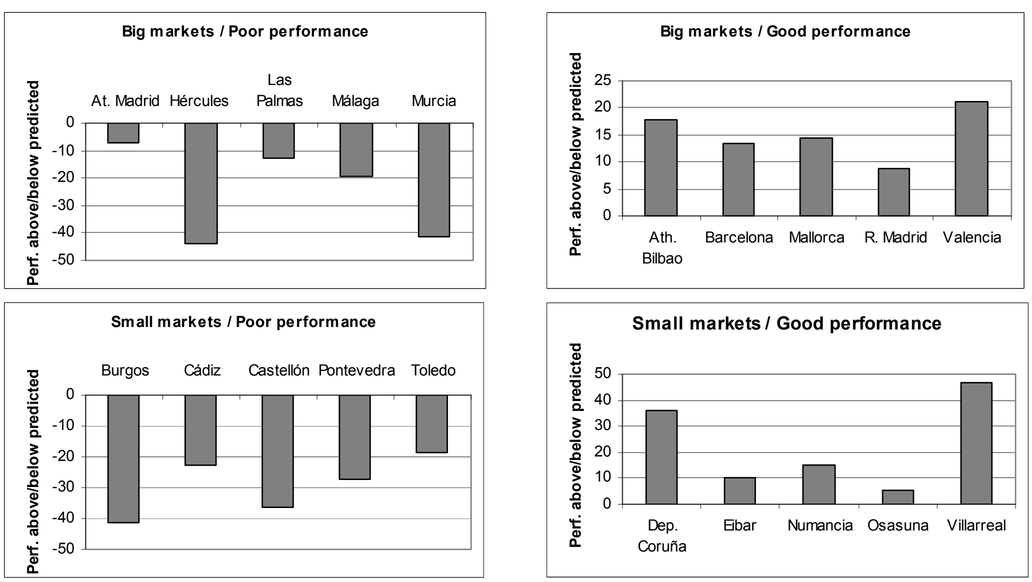
**Figure 1: Scatterplot of the *predicted performance* (PP) *vs. performance above/below the predicted level* (PABP)**



From this perspective, it is possible to measure how well teams performed above or below the levels predicted by local market size. In principle, this kind of comparison reveals that without their big-market advantage, teams located in the largest markets (Real Madrid, Barcelona and Valencia) would exhibit a substantial reduction in their playing success, and that teams with the best performance above/below predicted do not serve the largest markets (such as Deportivo A Coruña, Real Sociedad, Celta, and Villarreal). In particular, a very important result is that the performance of the teams serving the largest markets in Spain (Barcelona and Real Madrid) is clearly determined by their market size.

More specifically, we can use deviations between observed and predicted performance to distinguish at least four groups of teams (Figure 2). The first is a group comprised of big-market teams with a relatively poor performance (for instance, Atlético de Madrid, Hércules, Las Palmas, Málaga and Murcia); the second is a group of big-market teams with a relatively good performance (such as Athletic Bilbao, FC Barcelona, Real Madrid, Valencia and Mallorca); the third is a group of small-market teams with a relatively poor performance that is not attributable to market size (for instance, Burgos, Cádiz, Castellón, Pontevedra and Toledo); and finally, a group of small-market teams with a relatively good performance (such as Deportivo, Eibar, Numancia, Osasuna and Villarreal).

**Figure 2: Performance above (+) or below (-) predicted of the Spanish football teams (Model 1, *adjust. town population*): some illustrative examples**



Furthermore, data on predicted performance allows us to estimate some threshold values, namely the minimum values required to guarantee a particular long-term level of success if sporting performance were determined solely by the team's local market size. Table 4 shows the local market size threshold value for various levels of success calculated from the results of Model 1, using the specification for the variable *adjusted town population*.[11]

**Table 4: Minimum local market sizes necessary to guarantee different playing success levels**

| Playing success level | Local market size threshold value (adjusted town population, thousands) |
|---|---|
| UEFA Champions League appearance | 3,751.83 |
| UEFA Cup appearance | 2,100.65 |
| Keep on First Division | 437.03 |

The information provided by these values can be a useful reference in order to: (i) establish realistic sports objectives (both by clubs and fans), and (ii) evaluate whether the level of competitive balance may be excessively harmful or, alternatively, if the advantage of big cities is too large. Insofar as we can identify teams located in areas with a market size below those thresholds, but which are nonetheless capable of overcoming their predicted levels of success, it would seem that their disadvantage is not insurmountable. Therefore, according to these threshold values, it would be reasonable to assume that each club sets its expectations on sporting success at a different level, so that demand and fan satisfaction is not only a function of what is done, but also of what is expected of each team.

## V. CONCLUDING REMARKS

The empirical evidence derived from our research allows us to gain a greater understanding of the link between local market size and the playing success of Spanish soccer teams. More specifically, the main findings can be summarized as follows.

Firstly, big-market teams have a clear and meaningful advantage over small-market teams. Moreover, while geographical proximity is no longer a prerequisite for watching a soccer game or for purchasing a team product, the influence of local market size on sporting performance seems to be highly significant.

[11] The threshold values were calculated for $y = 0.93$, $y = 0.88$ and $y = 0.61$, respectively, which are the lowest *playing success indices* required in each case.

Secondly, in the Spanish league the advantage of big-market teams results in a substantial dominance by a very small group of clubs. However, this advantage is not insurmountable, and does not prevent teams that are not located in areas with the smallest market size from at least having regular opportunities for success.

Thirdly, the ranking of teams ordered by *observed performance above/below predicted performance* reveals that is very likely that specific clubs experience prolonged playing success below or above the levels predicted by their local market size. This result could explain the fact that despite major imbalances reflected by standard indicators of sporting success, fans in Spain do not lose interest in the competition and, consequently, the aggregate demand is not adversely affected.

Finally, the approach developed in this article allows us to address a number of important policy implications. In particular, our results support the idea that a professional league may benefit from keeping population size differences between constituent teams as small as possible. Obviously, a strategy of this kind is not feasible in Spain or in other national leagues considered individually, as there are not enough large local markets. Furthermore, the econometric modeling and the evidence obtained provide a rational basis for League Planners when designing soccer leagues (i.e. the optimal number and location of constituent teams), and specifically for discussing the idea of promoting the creation of a cross-national league in Europe (the European Superleague).

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